

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data | Data: Ember (2026); Energy Institute (2025)

Fossil Fuels: 75% of Global GHG Emissions

5M+

premature deaths per year

from air pollution caused by fossil fuel combustion

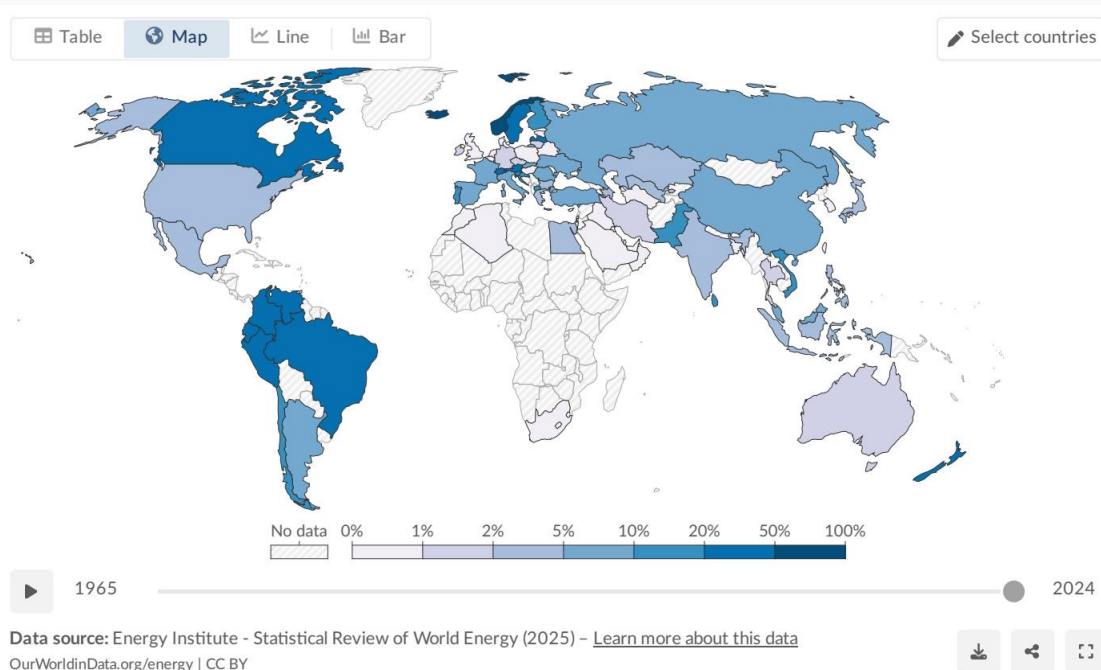
Since the Industrial Revolution, fossil fuels have dominated global energy systems, driving both climate change and severe local air pollution.

The Energy Transition Imperative

- Three-quarters of global greenhouse gas emissions come from burning fossil fuels for energy
- To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources
- Renewable technologies include solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources

Renewables Now Supply ~14% of Global Primary Energy

Primary energy encompasses electricity, transport, and heating combined (2024 data, substitution method).



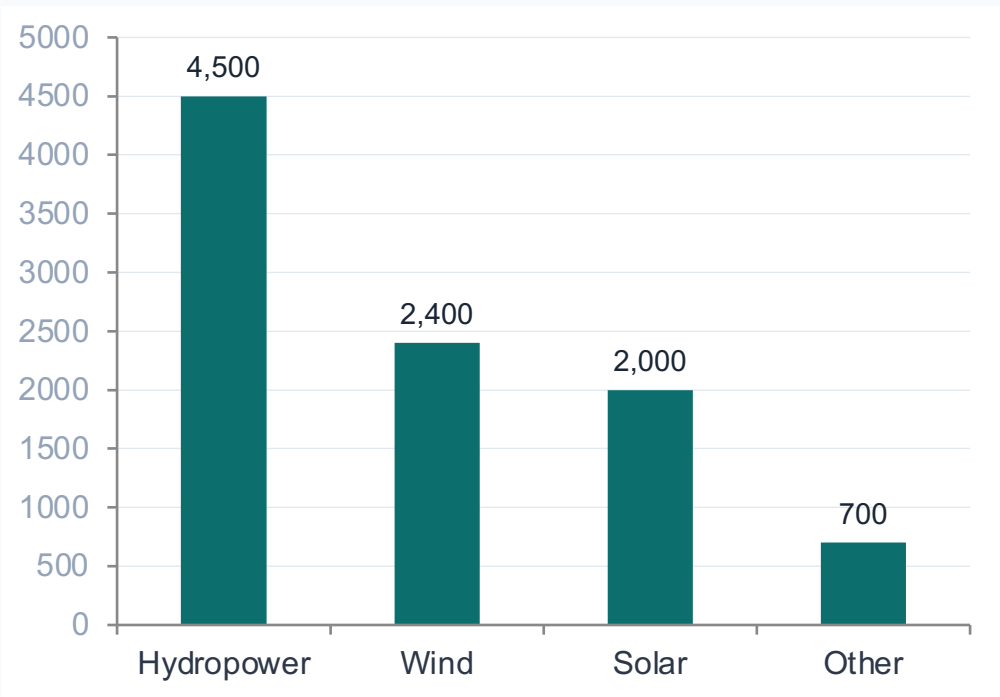
~1/7

of primary energy
is renewable

Transport and heating remain heavily
reliant on oil and gas, keeping the
renewable share lower than in
electricity.

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global renewable electricity generation by technology (2024 data)



Technology Breakdown

Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

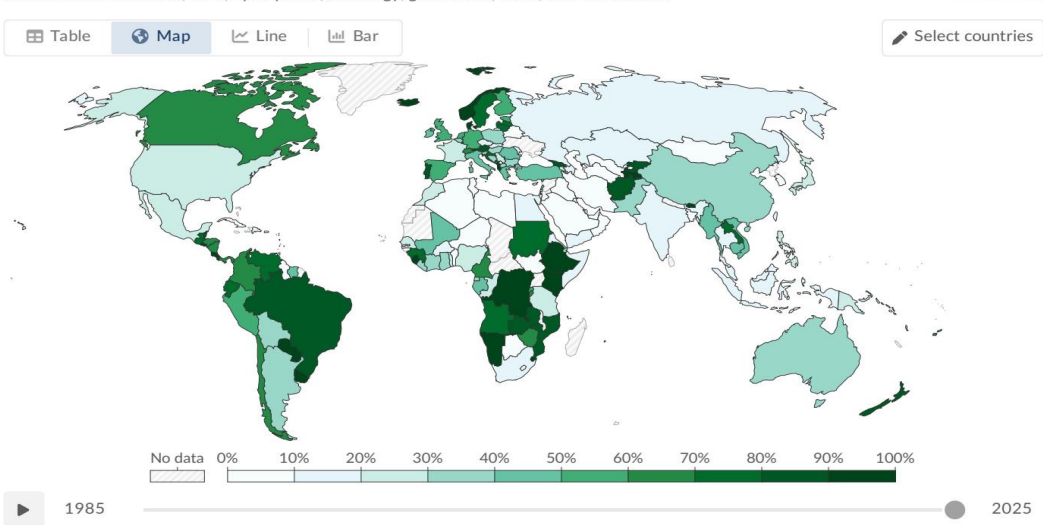
Wind + Solar combined now nearly match hydropower's output

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity from renewables by country (2025 data)

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

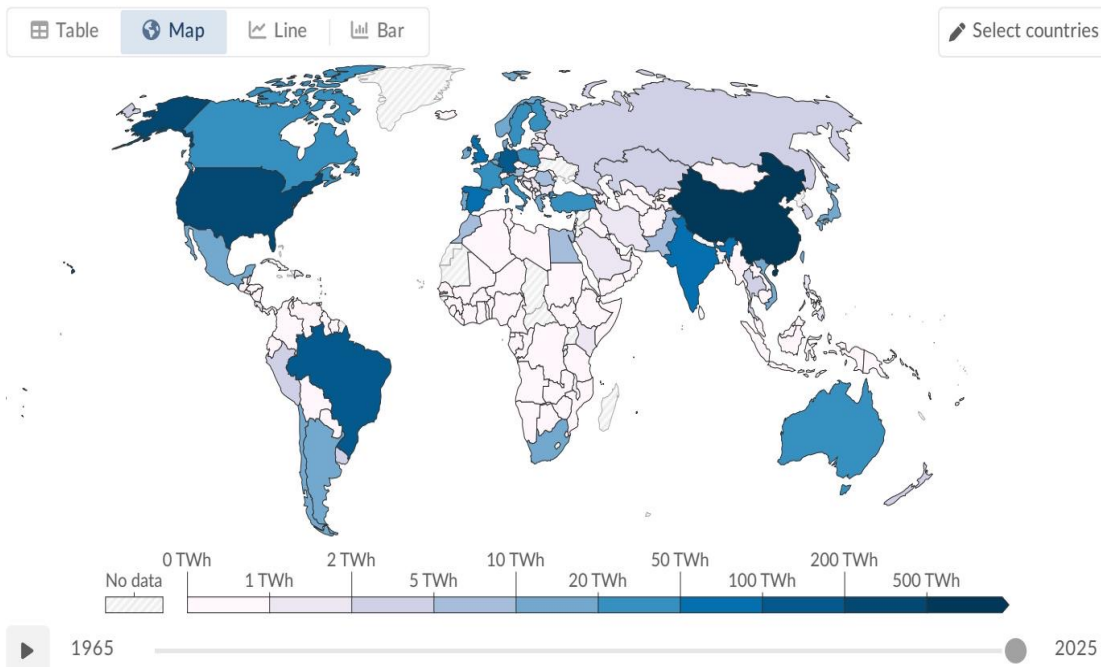
Renewable Electricity Share

Country/Region	Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

World average: ~30%

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower generation by country (2025 data)



Key Facts

~50% of all renewable electricity

Top producers:

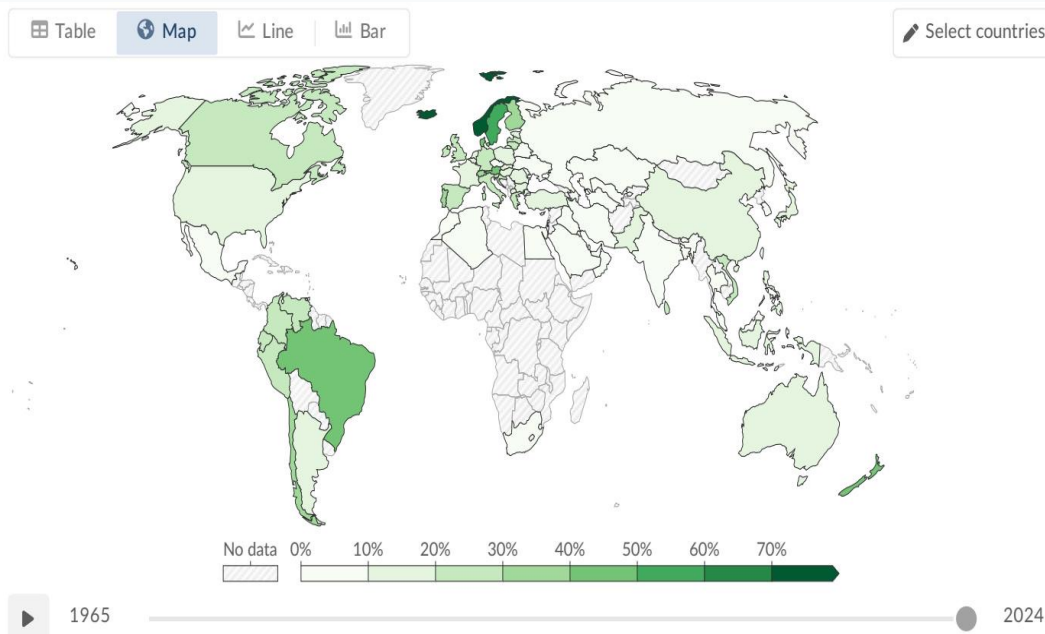
China

- Brazil
- Canada

Hydropower has been generating electricity at scale for over a century

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind power generation by country (2025 data)



Wind

~2,400 TWh

~25% of renewable electricity

Solar

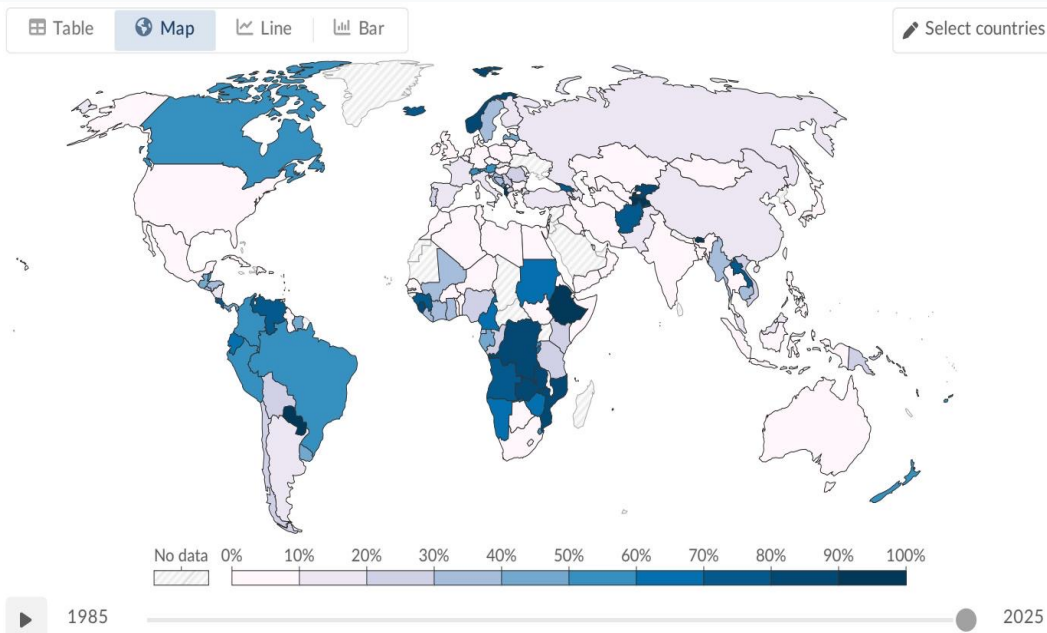
~2,000 TWh

~21% of renewable electricity

Fastest-growing source with dramatic growth since 2010, driven by falling costs and policy support

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia Leads

Share of electricity from hydropower by country (2025 data)



Regional Leaders

- Scandinavia & Brazil: 65-98%
 - Abundant hydropower resources
- Europe (Denmark, Germany): Heavy wind & solar investment

Regions Behind

- Sub-Saharan Africa: <10%
 - Minimal renewable infrastructure
- Middle East: Almost entirely fossil-fuel dependent

Key Takeaways

~14%

of primary energy

and ~30% of electricity now from
renewables

50%

hydropower dominates

but wind & solar are catching up fast

Regional

disparities persist

Scandinavia leads; MENA & Africa lag

The transition must accelerate. Transport and heating lag far behind electricity in the renewable transition, keeping the overall primary energy share much lower than the electricity share. To meet climate targets, wind and solar growth must continue to rapidly scale.